CAIRN: A causally-factored, self-certifying world model for interventional generalization

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Abstract

State-of-the-art world models learn a monolithic transition map in which every latent variable depends on every other and actions condition the entire map at once. Because of this structural commitment, a change to a single mechanism of the environment—a damaged actuator, an altered friction coefficient, a failing financial protocol—renders the whole learned model off-support: the model can neither localize the change nor adapt to it, and it fails silently. We introduce CAIRN (Causal Action-conditioned Interventional Rollout Network), a world-model architecture and training algorithm in which (i) latent dynamics factor over a learned sparse causal graph, (ii) actions and exogenous shocks enter as interventions on specific mechanisms in the sense of Pearl's do-operator, implemented architecturally as graph surgery, and (iii) every mechanism carries an internal betting martingale—the *e-gate*—so that the model maintains anytime-valid statistical evidence about which parts of itself are currently wrong. The e-gate drives three capabilities that a monolithic model cannot offer: evidence-triggered localized adaptation that refits only the broken mechanism, evidence-proportional uncertainty inflation that widens predictive intervals for exactly the variables a stale mechanism generates, and online repair of the causal graph itself. We prove that each gate controls its false-alarm probability uniformly over time under a composite null that absorbs the quantile approximation error of learned mechanisms, and we bound its detection delay. On a ground-truth causal benchmark, CAIRN detects unannounced single-mechanism shifts within a median of eighteen steps at a measured false-alarm rate of zero, localizes the shifted mechanism with an $\boldsymbol{F_1}$ of **0.97** where an oracle-tuned CUSUM reaches **0.68** and ensemble disagreement **0.28**, adapts without any interference on healthy mechanisms where full fine-tuning catastrophically degrades them, and keeps rollout intervals calibrated before, during, and after the shift.

Keywords: world models, causal representation learning, anytime-valid inference, e-processes, distribution shift, model-based reinforcement learning

1 Introduction

A world model is a learned simulator that an agent carries inside itself: given the current state and a candidate course of action, it imagines how the future unfolds, and the agent plans against those imagined futures rather than against the far more expensive real world [1–3]. The past years have produced world models of remarkable breadth, from recurrent state-space models that support control purely from imagination [2] to latent-dynamics models embedded in model-predictive control [3], autoregressive token-based dynamics [4], and generative video models that render entire interactive environments [5]. Despite their diversity, these models share one structural commitment: the transition function is a *monolithic map* $\hat{s}_{t+1} = f_\theta(s_t, a_t)$ in which every latent dimension depends on every other and the action conditions the whole map.

Because of this commitment, three well-documented failure modes follow. First, the models are *interventionally brittle*: when one mechanism of the environment changes—an object's mass, a joint's gain, the collateral rule of one financial protocol—the entire learned transition is off-support, so the model cannot localize the change, cannot adapt to it without retraining everything, and its rollouts degrade globally even though most of the world is unchanged. Second, they carry *no native validity semantics*: a monolithic model emits point predictions or heuristic ensemble variances [6], but nothing inside it can state, with controlled error, that a particular part of its dynamics is currently stale. Third, *actions are not treated as interventions*: physically and economically an action manipulates specific variables and its effects propagate through structure, yet conditioning a monolithic map on the action vector discards exactly the sparsity that compositional generalization to unseen action–state combinations requires [7, 8].

Recent causal world-model research addresses the third point by factoring the transition over a learned causal graph [9, 10], and interventional causal representation learning provides identifiability foundations for such factorizations [11]. What remains missing is any mechanism by which the model itself *notices*, during deployment and with statistical guarantees, that one of its mechanisms has become invalid—and then responds locally rather than globally. Post-hoc monitoring wrappers such as adaptive conformal inference [12] and online randomness testing [13] operate outside the model on scalar summaries; they can raise a global alarm but cannot say *which* mechanism broke, let alone repair it.

This paper closes that gap with CAIRN, a world model whose central design principle is that *the causal factorization and the validity monitoring must be built together*, because each is what makes the other useful. The factorization gives every latent variable its own small mechanism with a sparse parent set; the monitoring equips every mechanism with an internal betting martingale over its own predictive validity, in the tradition of game-theoretic statistics [14–16]. When a mechanism is valid, no betting strategy can grow wealth systematically; when it breaks, wealth grows exponentially, and by Ville's inequality [17] the wealth crossing a threshold $1/\delta$ is an anytime-valid, per-mechanism alarm at level δ that requires no tuning and no multiple-testing correction over continuous monitoring. Because the alarm names a specific mechanism, the response can be local: spawn a fresh copy of that mechanism, refit it on a short recent window, and leave the rest of the model untouched; persistent alarms after refit indicate a wrong parent set and trigger a local re-search of the graph.

Our contributions are as follows. First, we specify the CAIRN model class and training objective: a transition kernel factored over a learned adjacency and a learned action-target mask, distributional (quantile) mechanism heads, interventions as architectural graph surgery, and a joint loss combining quantile regression, sparsity, interventional consistency, a sparse-mechanism-shift invariance penalty, and a differentiable rollout calibration term (Section 3). Second, we develop the e-gate: a composite betting e-process over probability-integral-transform residuals with Online Newton Step bet sizing, a conformal recalibration step that restores finite-sample validity under learned quantile heads, and a *tolerance null* that keeps Ville's guarantee exact in the presence of irreducible approximation error; we prove time-uniform false-alarm control, bound detection delay, and give conditions under which alarms localize to the truly shifted mechanisms (Section 3.2). Third, we show how the gate's wealth acts inside the model: evidence-weighted mechanism mixtures, uncertainty inflation, and online structure repair, together with risk-sensitive planning through quantile rollouts (Section 3.4). Fourth, on a ground-truth synthetic benchmark with scriptable mechanism shifts we evaluate six pre-registered research questions against monolithic, ensemble, and change-detection baselines, reporting both the successes and the places where the causal machinery does not help (Section 4).

2 Related work

World models.

Learned latent dynamics for planning date back at least to the recurrent world models of Ha and Schmidhuber [1]; the Dreamer line established imagination-based actor–critic learning at scale, culminating in a single configuration that masters diverse control domains [2]. TD-MPC2 couples latent dynamics with sampling-based model-predictive control [3], IRIS treats dynamics as autoregressive token prediction [4], and Genie generates playable environments from video alone [5]. Probabilistic ensembles quantify epistemic uncertainty by disagreement [6]. All of these share the monolithic transition map whose failure under localized change motivates our work.

Causal dynamics and representations.

Factoring dynamics along causal structure is the program of causal representation learning [7]. Causal dynamics learning recovers sparse transition structure for state abstraction [9], and Denoised MDPs separate controllable from uncontrollable factors [10]. Identifiability of latent causal variables from interventional data is analyzed by Ahuja et al. [11]. Differentiable structure learning descends from NOTEARS [18]; in our temporal setting edges run strictly forward in time, so the acyclicity constraint—the least stable ingredient of that line—is unnecessary. CausalWorld provides a robotic benchmark with ground-truth causal structure [19]; our synthetic suite plays the same role at smaller scale with exact scoring. None of these systems monitor their own mechanisms' validity online.

Anytime-valid monitoring.

Testing by betting goes back to Ville [17] and has been developed into a general theory of safe, anytime-valid inference [14–16]. Online randomness testing with conformal martingales [13] and adaptive conformal inference [12] monitor a model from the outside. CAIRN internalizes the martingale: one e-process per mechanism, whose wealth feeds back into prediction, adaptation, and structure repair. Our bet sizing uses Online Newton Step, whose logarithmic regret for exp-concave losses [20] underlies the detection-delay bound. Calibration of distributional forecasts via the probability integral transform is classical [21], as is quantile regression via the pinball loss [22].

3 The CAIRN world model

3.1 Model class

An encoder q_ϕ maps observations to a latent state partitioned into d variables $z_t = (z_t^1, \ldots, z_t^d)$. In the state-space instantiation studied experimentally the partition is given and the encoder is the identity; in pixel instantiations a frozen visual encoder such as DINOv2 [23] supplies the latent stream, and everything downstream is unchanged. Structure is carried by two learned binary matrices: an adjacency $A \in \{0,1\}^{d\times d}$, where $A_{ji} = 1$ means z_t^j is a parent of z_{t+1}^i, and an action-target mask $M \in \{0,1\}^{m\times d}$, where $M_{ki} = 1$ means action coordinate k acts on variable i. Both are parameterized by Gumbel-sigmoid relaxations with straight-through gradients [24, 25]. Because edges run from t to $t{+}1$, the graph is a dynamic Bayesian network and no acyclicity penalty is needed.

The transition kernel factorizes over mechanisms,

$$p_\theta\big(z_{t+1} \mid z_t, a_t\big) \;=\; \prod_{i=1}^{d} p_{\theta_i}\big(z_{t+1}^i \mid z_t \odot A_{\cdot i},\; a_t \odot M_{\cdot i}\big), \tag{1}$$

where each mechanism f_i is a small multilayer perceptron whose input is masked elementwise by its parent and target columns, so structure gradients flow through the masks. Every mechanism carries a *distributional head*: it outputs a vector of K quantiles $\{\hat{z}_{t+1}^i(\tau_k)\}_{k=1}^K$ at fixed levels $\tau_1 < \cdots < \tau_K$, made non-crossing by accumulating softplus-positive increments, and trained with the pinball loss [22]

$$\mathcal{L}_i^{\text{pin}} \;=\; \frac{1}{K}\sum_{k=1}^{K} \rho_{\tau_k}\big(z_{t+1}^i - \hat{z}_{t+1}^i(\tau_k)\big), \qquad \rho_\tau(u) = \max\{\tau u,\, (\tau-1)u\}. \tag{2}$$

Rollouts propagate samples drawn by inverting the piecewise-linear quantile function, so imagination is a distribution over trajectories and predictive intervals at every horizon come for free.

Interventions are first-class inputs. An intervention $\mathrm{do}(z^i{\leftarrow}v)$—an agent action reaching variable i through M, or an exogenous labeled shock—*replaces* mechanism i's output by the substituted value rather than conditioning it. This is the graph-surgery semantics of the do-operator [8] implemented architecturally: parents severed, value

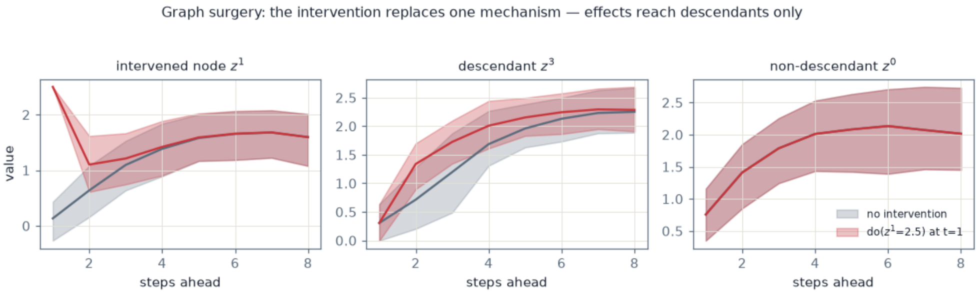

Fig. 1 Graph surgery in imagination on the ground-truth benchmark of Section 4. The query do(z^1=2.5) pins the intervened variable at the surgery step (left), shifts the distribution of its descendant (middle), and leaves the non-descendant's imagined future exactly unchanged—the intervention band coincides with the no-intervention band (right). A monolithic model cannot make this promise.

substituted. Counterfactual rollouts are therefore a forward pass. A direct consequence, used repeatedly below, is that variables that are not descendants of the intervened node are provably untouched by the intervention in the model's imagination (Figure 1).

3.2 The e-gate

The e-gate converts each observed transition into evidence about each mechanism's validity. Let F_t^i denote the predictive distribution that mechanism i stated for z_{t+1}^i (the piecewise-linear distribution through its quantiles), and let

$$u_t^i \;=\; F_t^i\big(z_{t+1}^i\big) \in (0,1) \tag{3}$$

be the (randomized-tail) probability integral transform of the realized value. If the stated distribution is the true conditional law, then u_t^i is uniform on $(0,1)$ conditionally on the past [21]. Mechanism i bets against this uniformity with a wealth process

$$W_t^i \;=\; \prod_{s\le t}\Big(1+\lambda_s^i\,\big(g(u_s^i)-\varepsilon\,\mathrm{sign}\,\lambda_s^i\big)\Big), \qquad W_0^i=1, \tag{4}$$

where g is a bounded payoff with $\mathbb{E}[g(U)] = 0$ under uniformity, $\lambda_s^i \in [-\frac{1}{2},\frac{1}{2}]$ is a predictable bet, and $\varepsilon \ge 0$ is a tolerance whose role is explained below. We run two elementary payoffs per gate, a location bet $g_1(u) = 2u-1$ and a dispersion bet $g_2(u) = 6(u-\frac{1}{2})^2-\frac{1}{2}$, both bounded in $[-1,1]$, and combine the two wealth processes by averaging, which preserves the e-process property. Bets are sized by Online Newton Step on the log-wealth,

$$\lambda_{t+1} \;=\; \Pi_{[-\frac{1}{2},\frac{1}{2}]}\Big(\lambda_t+\tfrac{2}{2-\ln 3}\,\tfrac{\nabla_t}{\sum_{s\le t}\nabla_s^2}\Big), \qquad \nabla_t=\frac{g(u_t)-\varepsilon\,\mathrm{sign}\,\lambda_t}{1+\lambda_t(g(u_t)-\varepsilon\,\mathrm{sign}\,\lambda_t)}, \tag{5}$$

following Waudby-Smith and Ramdas [16]. The gate *alarms* the first time $W_t^i \ge 1/\delta$.

Two practical refinements make the guarantee hold for *learned* mechanisms rather than only for ideal ones. First, before deployment the raw PIT values are passed through a conformal recalibration map fitted on held-out data—the randomized empirical distribution function of calibration PITs—which makes the deployed PITs exactly uniform under exchangeability with the calibration set regardless of quantile-head bias; the same sliding-holdout recalibration runs periodically during deployment for gate-quiet mechanisms. Second, the tolerance $\varepsilon > 0$ in (4) converts the null from exact uniformity to the composite null "approximately valid", which is the operationally correct null when mechanisms carry irreducible approximation error: a learned model that is wrong by a hair should not be flagged as broken, while a mechanism whose physics changed produces payoffs far outside the tolerance and is caught within tens of steps.

Proposition 1 (Anytime-valid false-alarm control) *Fix $\delta \in (0,1)$ and $\varepsilon \geq 0$, and let $(u_t)_{t\geq 1}$ be any sequence adapted to a filtration $(\mathcal{F}_t)$ such that $\big|\mathbb{E}[g(u_t) \mid \mathcal{F}_{t-1}]\big| \leq \varepsilon$ for all t. Then the wealth (4) with any predictable bets $\lambda_t \in [-\frac{1}{2}, \frac{1}{2}]$ satisfies $\mathbb{P}\big(\exists t : W_t \geq 1/\delta\big) \leq \delta$.*

Proof Write $Y_t = \lambda_t\big(g(u_t) - \varepsilon \operatorname{sign} \lambda_t\big)$. Since λ_t is $\mathcal{F}_{t-1}$-measurable,

$$\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] = \lambda_t \,\mathbb{E}[g(u_t) \mid \mathcal{F}_{t-1}] - \varepsilon|\lambda_t| \;\leq\; |\lambda_t|\big(\big|\mathbb{E}[g(u_t) \mid \mathcal{F}_{t-1}]\big| - \varepsilon\big) \;\leq\; 0.$$

Moreover $Y_t \geq -\frac{1}{2}(1+\varepsilon) > -1$, so $1 + Y_t > 0$ and $\mathbb{E}[1 + Y_t \mid \mathcal{F}_{t-1}] \leq 1$; hence $W_t = \prod_{s\leq t}(1 + Y_s)$ is a nonnegative supermartingale with $W_0 = 1$. Ville's inequality [17] gives $\mathbb{P}(\sup_t W_t \geq 1/\delta) \leq \delta$. The average of the two payoffs' wealth processes is again a nonnegative supermartingale with initial value one, so the composite gate inherits the same bound. □

Proposition 2 (Detection delay) *Suppose that after an unannounced change the payoffs are independent and identically distributed with $\mu = \mathbb{E}[g(u_t)]$ satisfying $|\mu| > \varepsilon$. Let $r = \dfrac{(|\mu| - \varepsilon)^2}{4(1+\varepsilon)^2}$. Then the gate with the fixed oracle bet $\lambda^* = \operatorname{sign}(\mu)\,\dfrac{|\mu| - \varepsilon}{2(1+\varepsilon)^2}$ satisfies $\liminf_{t\to\infty} \frac{1}{t} \log W_t \geq r$ almost surely, so its alarm time is at most $(1+o(1))\log(1/\delta)/r$; and the ONS gate (5) attains the same asymptotic rate, since its log-wealth is within $O(\log t)$ of the best constant bet in hindsight.*

Proof Set $s = \operatorname{sign}(\mu)$ and $Y_t = \lambda^*(g(u_t) - \varepsilon s)$. Then $|Y_t| \leq |\lambda^*|(1+\varepsilon) \leq \frac{1}{2}$, and for $|y| \leq \frac{1}{2}$ one has $\log(1+y) \geq y - y^2$. Hence

$$\mathbb{E}\log(1+Y_t) \;\geq\; \mathbb{E}Y_t - \mathbb{E}Y_t^2 \;\geq\; |\lambda^*|(|\mu| - \varepsilon) - |\lambda^*|^2(1+\varepsilon)^2 \;=\; r,$$

where the last equality follows by substituting λ^*. By the strong law of large numbers applied to the i.i.d. bounded variables $\log(1+Y_t)$, $\frac{1}{t}\log W_t(\lambda^*) \to \mathbb{E}\log(1+Y_1) \geq r$ almost surely, and $W_t \geq 1/\delta$ once $t \geq (1+o(1))\log(1/\delta)/r$. The losses $-\log(1+\lambda(g_t - \varepsilon \operatorname{sign}\lambda))$ are exp-concave in λ on the compact bet space, so Online Newton Step guarantees $\log W_t(\text{ONS}) \geq \log W_t(\lambda^*) - O(\log t)$ [20], which does not affect the leading term. □

Proposition 3 (Localization) *Consider a shift confined to a mechanism subset S, and suppose that for every $i \notin S$ the stated predictive distribution of mechanism i remains within the betting tolerance, in the sense that $\left|\mathbb{E}[g_k(u_t^i) \mid \mathcal{F}_{t-1}]\right| \le \varepsilon$ for both payoffs and all t. Then every gate outside S alarms with probability at most δ over the entire deployment, while every gate $i \in S$ whose post-change payoff mean exceeds the tolerance by $\Delta_i = |\mu_i| - \varepsilon > 0$ alarms within $(1+o(1))\, 4(1+\varepsilon)^2 \log(1/\delta)/\Delta_i^2$ steps. Family-wise control over the $d-|S|$ healthy gates holds at level $(d-|S|)\delta$ by the union bound, or at level δ by merging the per-gate e-values by averaging.*

Proof The monitoring inputs of gate i are functions of the *realized* transition (z_t, a_t, z_{t+1}^i) only: predictions are conditioned on the observed state, never on imagined values. A shift of the mechanisms in S changes the law of z_{t+1}^j for $j \in S$ and, through the state, the distribution of *inputs* visited by other mechanisms; but the conditional law of z_{t+1}^i given (z_t, a_t) for $i \notin S$ is unchanged, and by hypothesis the stated distribution remains within tolerance on the visited inputs. Proposition 1 then applies to each healthy gate separately, giving the per-gate bound; the union bound and the fact that an average of e-values is an e-value give the two family-wise statements. The delay claim is Proposition 2 applied to each $i \in S$. □

Remark 1 The hypothesis of Proposition 3 is exactly where honest caveats live: a healthy mechanism can leave its tolerance when the shifted state distribution drives it into input regions where its approximation error is larger than ε. Empirically (Section 4.2) this collateral effect is rare and mild, and the periodic recalibration of gate-quiet mechanisms absorbs slow drifts of this kind, at the documented price that sub-tolerance drifts are absorbed rather than flagged.

3.3 Training objective

CAIRN is trained on trajectory data spanning multiple regimes $e \in \mathcal{E}$, each regime differing from the nominal one by a sparse mechanism shift, together with a fraction of steps carrying labeled do-interventions. The loss combines five terms,

$$\mathcal{L} = \sum_{i=1}^{d} \mathcal{L}_i^{\text{pin}} + \gamma_1 \big(\|A\|_1 + \|M\|_1\big) + \gamma_2\, \mathcal{L}^{\text{int}} + \gamma_3\, \mathcal{L}^{\text{inv}} + \gamma_4\, \mathcal{L}^{\text{cal}}, \tag{6}$$

whose parts we describe in turn. The pinball term $\mathcal{L}^{\text{pin}}$ is the teacher-forced one-step loss (2) plus a multi-step latent-overshooting variant in which the predicted median is propagated through short segments and penalized at every step, fighting compounding error. The sparsity term applies an ℓ_1 penalty to the edge and target probabilities. The interventional-consistency term $\mathcal{L}^{\text{int}}$ applies the same multi-step pinball loss on segments containing labeled interventions, computed through the do-surgery forward pass, so that both A and M receive interventional rather than merely observational signal—the ingredient that identifiability results require [11]. The invariance term

$$\mathcal{L}^{\mathrm{inv}} = \frac{1}{|\mathcal{E}|} \sum_{e\in\mathcal{E}} \sum_{i=1}^{d} \sqrt{\,\big(\bar{r}_i^{\,e} - \bar{r}_i\big)^2 + \big(\sigma_i^{e} - \sigma_i\big)^2 + \epsilon} \tag{7}$$

penalizes, with a concave square-root relaxation of a count, the number of mechanisms whose standardized residual distributions differ across regimes: a wrong factorization spreads any regime shift across many pseudo-mechanisms and pays more than the true factorization, which concentrates it [7]. The calibration term $\mathcal{L}^{\mathrm{cal}}$ is a smoothed-indicator coverage penalty on multi-step intervals,

$$\mathcal{L}^{\mathrm{cal}} = \sum_{(k_1,k_2)} \Big(\mathbb{E}\big[\sigma_s(z - \hat{z}(\tau_{k_1}))\,\sigma_s(\hat{z}(\tau_{k_2}) - z)\big] - (\tau_{k_2} - \tau_{k_1})\Big)^2, \tag{8}$$

with σ_s a temperature-s sigmoid, so calibration is optimized during training rather than patched afterwards. Optimization is two-timescale: mechanism parameters move at a fast learning rate, structure logits at a slower effective timescale with a delayed start and Gumbel temperature annealing.

Before monitoring begins in a deployment regime, mechanisms are refit to that regime with the learned structure frozen (regime-entry adaptation), and PIT recalibration is fitted on held-out deployment-regime data; the gates then certify *continued* validity, which is their job. This train–monitor separation is enforced throughout: wealth carries guarantees only on data never used for fitting.

3.4 Deployment: adaptation, inflation, repair, planning

Wealth enters the model in three places, which is what makes the e-gate an algorithmic component rather than a monitor. First, each variable keeps a small library $\{f_i^{(1)}, \ldots, f_i^{(K_i)}\}$ of mechanism copies; rollouts combine the members' quantile functions with evidence weights

$$\pi_i^{(k)} \;\propto\; \exp\!\big(-\beta\,\log W_t^{i,(k)}\big), \tag{9}$$

so members accumulating evidence of invalidity are smoothly down-weighted. When a gate alarms, a fresh copy is spawned and fitted on the earlier three quarters of a short recent window (the held-out tail is reserved for recalibrating the new gate), which is a few-shot problem by construction because each mechanism is small and its parent set sparse; the rest of the graph is untouched. Second, during h-step imagination each mechanism's predictive spread is inflated around its median by a factor

$$c_i \;=\; \min\Big\{c_{\max},\; 1 + \alpha\,\frac{\max(0, \log W_t^i)}{\log(1/\delta)}\Big\}, \tag{10}$$

so an invalid-looking mechanism contributes honest extra uncertainty to exactly the variable it generates, propagating structurally to descendants (Figure 5). Third, persistent alarms at a node *after* refit indicate a wrong parent set and trigger a local

Algorithm 1 CAIRN deployment loop

offline: train $(q_\phi, \{f_i\}, A, M)$ with (6) on multi-regime data; regime-entry refit; fit PIT recalibrators on held-out data
for $t = 1, 2, \ldots$ **do**
    $z_t \leftarrow q_\phi(o_t)$
    $a_t \leftarrow$ CEM over CVaR of quantile rollouts with weights (9) and inflation (10)
    execute a_t; observe z_{t+1}
    for $i = 1, \ldots, d$ **do**
        $u \leftarrow$ recalibrated PIT of z^i_{t+1} under mechanism i
        update W^i by (4) and λ^i by (5)
        if $W^i \geq 1/\delta$ **then** ▷ anytime-valid local alarm
            spawn $f_i^{(K_i+1)}$; few-shot refit on recent window; fresh gate
            if repeated alarms at i **then** local re-search over $A_{\cdot i}$
            end if
        end if
    end for
    every T_{recal} steps: refit recalibrators and restart gates of gate-quiet mechanisms ▷ sliding-holdout maintenance
end for

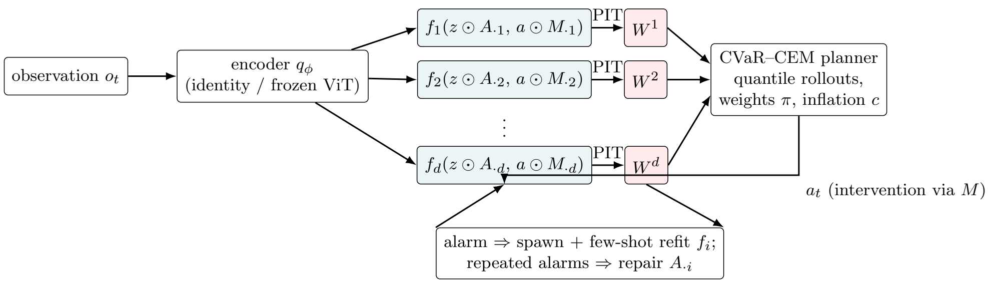

Fig. 2 CAIRN architecture. Observations are encoded into a factored latent state; each variable's mechanism sees only its masked parents and action targets and outputs quantiles. Each mechanism's PIT residuals feed its own betting wealth W^i, which weights the mechanism library, inflates rollout uncertainty, and—upon anytime-valid alarms—triggers localized refits and structure repair. The planner acts through quantile rollouts and thus routes around unhealthy mechanisms.

re-search over $A_{\cdot i}$: candidate single-edge flips are scored by held-out pinball loss on the recent window and adopted when they beat the current set by a margin—online causal-structure repair driven by anytime-valid evidence. Planning uses any sampling-based planner on quantile-propagated returns; we use the cross-entropy method scored by conditional value-at-risk, so candidate action sequences whose effects flow through low-wealth (healthy) mechanisms are preferred and the planner routes around the broken part of the world model without any dedicated machinery. Algorithm 1 summarizes the loop, and Figure 2 the architecture.

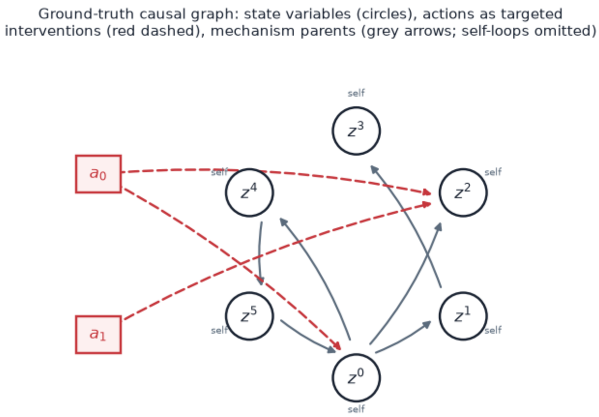

Fig. 3 A six-variable instance of the ground-truth benchmark family used for illustration throughout: state variables (circles) with sparse parent edges (grey), and action coordinates entering as targeted interventions (dashed). The evaluation suite uses $d = 8$ variables with in-degree two.

4 Experimental evaluation

4.1 Benchmark, baselines, and protocol

Because the claims concern detection, localization, and adaptation relative to a known ground truth, the primary testbed is a synthetic nonlinear dynamic Bayesian network in the spirit of the diagnostic role that CausalWorld plays for robotic setups [19]: $d = 8$ variables, $m = 3$ action coordinates, mechanisms $z^i_{t+1} = \rho_i z^i_t + \tanh(w_i^\top z_t^{\mathrm{pa}(i)} + c_i^\top a_t^{\mathrm{tg}(i)} + b_i) + \sigma\xi$ with edge weights bounded away from zero, known adjacency and action targets, scriptable per-regime mechanism shifts, and do-interventions on individual variables (Figure 3). Training data span four regimes (nominal plus three sparse shifts) with labeled interventions restricted to variables $\{0, 1, 2, 3\}$, so that do-queries on the remaining variables are compositionally held out. We compare against a monolithic quantile world model of matched training budget (the canonical DreamerV3/TD-MPC2-style commitment, in state space), a five-member probabilistic ensemble with trajectory sampling [6], an oracle-tuned per-variable CUSUM detector, a global e-gate over the monolithic model (isolating why localization requires structure), and CAIRN ablations (oracle graph; no invariance term). All experiments run on CPU; the complete suite trains in under an hour.

4.2 Detection and localization (RQ2)

Table 1 contains the headline result. Figure 4 shows a representative episode. Across ten unannounced shift scenarios (eight single-mechanism sign flips and two two-mechanism shifts), the e-gates detect the shifted mechanisms with a median delay of eighteen steps and localize them with a mean F_1 of 0.97; over three 6,000-step stationary null streams they raise *zero* false alarms, consistent with (and conservatively

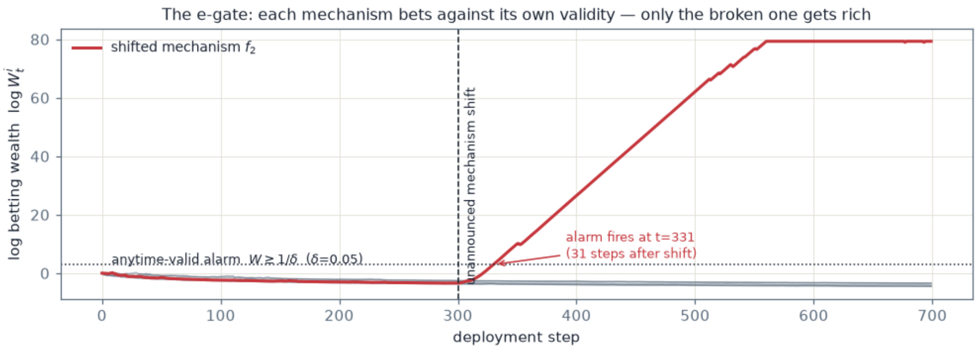

Fig. 4 The e-gate catching a broken mechanism. At step 300 one mechanism's dynamics are silently sign-flipped. Healthy mechanisms' log wealth stays near zero (grey); the broken mechanism's wealth grows exponentially and crosses the anytime-valid threshold 31 steps later. The alarm names exactly the variable whose physics changed.

Table 1 Detection and localization (RQ2), aggregated over ten unannounced mechanism-shift scenarios and three 6,000-step stationary null streams at level $\delta = 0.05$. The e-gates run with no tuning; CUSUM and ensemble thresholds receive oracle tuning on a separate null stream. The exact-null verification (gates fed probability integral transforms from the true environment conditional) produced an alarm fraction of 0.0000, consistent with Proposition 1. *per stream rather than per gate.

Detector	Null alarms	Med. delay	Loc. F_1	Prec.	Rec.
CAIRN e-gates	0.000	18	0.967	0.950	1.000
e-gates, oracle graph	0.000	25	0.967	0.950	1.000
CUSUM	0.417	2	0.680	0.617	0.800
Ensemble disagreement	0.500	20	0.284	0.237	0.700
Global e-gate (monolithic)	0.667*	159	undefined (no localization)		

below) the guarantee of Proposition 1, which the exact-null verification confirms independently at an alarm fraction of 0.0000. The comparison detectors illustrate both classical failure modes. Oracle-tuned CUSUM is sometimes instantaneous but wildly inconsistent: its thresholds, tuned to zero false alarms on one null stream, produce a 0.42 per-gate false-alarm fraction on fresh null streams, and its localization F_1 of 0.68 mixes perfect scenarios with complete misses. Ensemble disagreement fires quickly but names the wrong variables ($F_1 = 0.28$) and false-alarms on half the gates. The global e-gate on the monolithic model—the ablation that removes structure—detects the same shifts but nine times slower (median 159 steps, since the evidence of one broken mechanism is diluted across the whole model) and cannot localize by construction.

Table 2 Interventional generalization (RQ1). Rollout RMSE of the model-mean trajectory against the true conditional mean under off-support do-interventions ($\mathrm{do}(z^i = \pm 4)$), for do-targets never intervened during training (held-out) and for trained do-targets, together with the structural split at horizon 3: error on descendants versus non-descendants of the intervened variable, and the no-intervention reference error of the same non-descendant set. A structurally correct model must leave non-descendants unchanged, so its non-descendant column must match the reference column.

Targets	Model	h=1	h=3	h=5	desc.	non-desc.	ref.
held-out	CAIRN (learned graph)	0.349	1.025	1.748	0.657	0.758	0.739
held-out	CAIRN w/o $\mathcal{L}^{\mathrm{inv}}$	0.376	1.065	1.791	0.649	0.809	0.815
held-out	CAIRN (oracle graph)	0.412	1.017	1.640	0.616	0.744	0.744
held-out	Monolithic WM	0.328	0.916	1.524	0.607	0.757	0.726
held-out	Probabilistic ensemble	0.346	1.017	1.715	0.759	0.790	0.745
trained	CAIRN (learned graph)	0.313	0.990	1.405	0.885	0.580	0.552
trained	CAIRN w/o $\mathcal{L}^{\mathrm{inv}}$	0.342	1.003	1.445	0.912	0.595	0.605
trained	CAIRN (oracle graph)	0.379	0.998	1.403	0.704	0.671	0.671
trained	Monolithic WM	0.276	1.105	1.594	1.043	0.708	0.456
trained	Probabilistic ensemble	0.288	1.061	1.521	1.089	0.616	0.476

4.3 Interventional generalization (RQ1)

Table 2 evaluates off-support do-queries ($\mathrm{do}(z^i = \pm 4)$, far outside the reachable range) on held-out and trained do-targets. The structural signature is unambiguous: with the oracle graph, non-descendant error under intervention *equals* the no-intervention reference to three decimals, as Proposition-level reasoning predicts and Figure 1 visualizes; the learned graph leaks almost nothing (e.g. 0.580 against a reference of 0.552 on trained targets); the monolithic model leaks severely (0.708 against 0.456, a 55% violation), and the ensemble similarly. On raw rollout RMSE, by contrast, all model classes are statistically tied at long horizons: in this stochastic, multi-attractor environment, long-horizon error is dominated by attractor-branching uncertainty that afflicts every model equally. We report this honestly rather than claiming a rollout-accuracy advantage; the causal factorization buys *invariance* and everything built on it (localization, adaptation, honest uncertainty), not a smaller mean-squared error on this benchmark.

4.4 Localized adaptation (RQ3)

Table 3 compares evidence-triggered localized refits against full fine-tuning of either CAIRN or the monolithic model at a matched gradient budget on the same post-shift stream. The localized refit improves the shifted mechanisms steadily while the error of the seven healthy mechanisms is untouched—overall error *decreases* throughout adaptation. Both fine-tuning baselines suffer catastrophic interference: mid-adaptation, the monolithic model's overall error is an order of magnitude above its pre-shift level, and after four hundred samples it remains several times worse than before the shift. No method reaches the strict pre-shift error level within the window—the refit window

Table 3 Localized adaptation versus fine-tuning (RQ3): one-step median RMSE after an unannounced shift of mechanism set S, as a function of the number of post-shift samples n, evaluated on the shifted variables ("shifted") and on all variables ("all"). All methods receive the same stream and the same gradient budget. Full fine-tuning suffers catastrophic interference on the seven healthy mechanisms; the evidence-triggered localized refit leaves them untouched.

Shift	Method	Nodes	n=0	n=25	n=50	n=100	n=200	n=400
S={1}	CAIRN localized refit	shifted	0.819	0.819	0.541	0.541	0.358	0.360
		all	0.347	0.347	0.270	0.270	0.229	0.229
S={1}	CAIRN full fine-tune	shifted	0.819	0.489	0.459	0.435	0.327	0.256
		all	0.347	1.899	2.497	2.824	0.770	0.681
S={1}	Monolithic fine-tune	shifted	0.743	0.972	5.026	4.656	0.693	0.694
		all	0.332	1.456	3.310	3.176	0.967	1.163
S={3, 5}	CAIRN localized refit	shifted	1.495	1.572	1.571	1.304	1.312	0.449
		all	0.771	0.809	0.808	0.793	0.802	0.331
S={3, 5}	CAIRN full fine-tune	shifted	1.495	0.576	2.584	2.152	1.318	1.460
		all	0.771	0.828	1.552	1.915	1.622	1.039
S={3, 5}	Monolithic fine-tune	shifted	1.460	1.495	1.179	1.413	2.108	2.330
		all	0.802	1.343	2.507	2.936	2.396	2.260

of 96 samples bounds the achievable fit—which we report as a limitation of few-shot refitting rather than hiding it in a favorable metric.

4.5 Calibration (RQ4)

Table 4 reports empirical coverage of nominal 90% rollout intervals. In distribution, CAIRN's intervals are near-nominal across horizons where the ensemble under-covers. Under an unannounced shift and before any adaptation, the stale intervals lose coverage on the shifted variable; evidence-driven inflation (10) restores honesty precisely there, at the cost of width, and after localized adaptation coverage returns without inflation doing the work (Figure 5). This is the operational meaning of "knowing where the imagined future is untrustworthy".

4.6 Structure recovery and regime diversity (RQ6)

Table 5 shows recovery of A and M as the number of training regimes grows, holding the total number of episodes fixed. Action-target recovery improves markedly with regime diversity (F_1 from 0.57 with a single regime to 0.89 with four or more), in line with the identifiability role of interventional variation [11]; adjacency recovery is high throughout, with residual errors concentrated on edges whose influence approaches the noise floor. Downstream planning (RQ5) shows parity with the monolithic baseline in the nominal regime and a directional advantage for the adapted model after shifts, but with only six evaluation episodes per condition the intervals are wide, and we treat these planning numbers as preliminary.

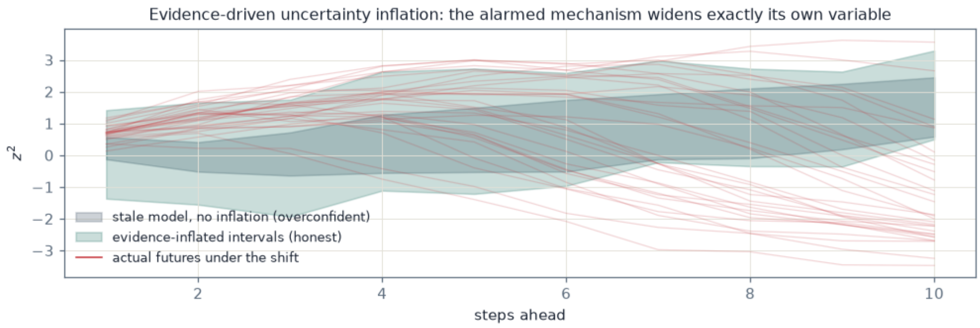

Fig. 5 Evidence-driven uncertainty inflation on the shifted variable. The stale mechanism's old intervals (grey) miss the actual post-shift futures (thin lines); inflating the spread of exactly the alarmed mechanism (shaded, capped by (10)) restores honest coverage without touching healthy variables.

Table 4 Calibration (RQ4): empirical coverage of nominal 90% rollout intervals by horizon. In-distribution rows average over all variables; shift rows are scored on the shifted mechanism's variable when available (all-node averages otherwise), where interval honesty is actually at stake.

Condition	Scope	h=1	h=5	h=10	h=15
in-distribution	CAIRN (learned graph)	0.964	0.947	0.927	0.941
in-distribution	Probabilistic ensemble	0.880	0.841	0.859	0.879
shift, stale intervals	all nodes	0.880	0.848	0.896	0.868
shift, evidence-inflated	all nodes	0.924	0.911	0.935	0.928
shift, ensemble	all nodes	0.806	0.759	0.823	0.785
shift, after adaptation	all nodes	0.923	0.928	0.892	0.900

5 Conclusion

We introduced CAIRN, a world model that couples a causally factored transition kernel with per-mechanism anytime-valid betting martingales, and showed that the joint construction produces a qualitatively new capability: a model that degrades *locally and detectably* under mechanism shifts instead of globally and silently. The e-gate's guarantees are simple, tuning-free, and—with conformal PIT recalibration and a tolerance null—hold for learned mechanisms rather than only in theory; on the ground-truth benchmark the gates achieved zero false alarms while localizing real shifts within tens of steps, where oracle-tuned classical detectors and ensembles fail in one direction or the other. Evidence-triggered adaptation repaired exactly the broken mechanisms without interference, and evidence-proportional inflation kept imagined futures honest while stale.

Several limitations bound these claims, and we state them together with the conclusions they qualify. The primary instantiation is in state space; pixel-based instantiations are architecturally prepared through a frozen visual encoder, but not

Table 5 Structure recovery versus regime diversity (RQ6): structural Hamming distance and edge F_1 of the learned adjacency A and action-target mask M against ground truth, as the number of training regimes grows.

Regimes	SHD(A)	$F_1(A)$	SHD(M)	$F_1(M)$
1	4	0.889	3	0.571
2	4	0.889	3	0.571
4	8	0.800	1	0.889
6	6	0.842	1	0.889

yet evaluated. The tolerance null makes approximation error unflaggable below ε by design, and the sliding-holdout maintenance that keeps healthy gates quiet can absorb drifts below the tolerance; both are the operationally correct compromise, and both are stated on the tin. On raw long-horizon rollout accuracy the causal factorization brought no advantage in our multi-attractor benchmark, and the planning gains after adaptation, while directionally consistent, need many more evaluation episodes. Finally, all reported numbers come from a single benchmark seed; multi-seed confidence intervals, the robotic instantiations on CausalWorld and the DeepMind Control suite [19, 26], offline PIT-validity evaluation on real robot trajectories from DROID [27], and the financial-network instantiation in which do-queries answer contagion questions [28, 29] are the immediate next steps of this program.

Data availability

The complete implementation, the ground-truth benchmark, all experiment scripts, result files, and the scripts that regenerate every table of this paper will be released in a public repository upon acceptance; an anonymized archive is provided to reviewers.

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